



Portfolio

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LinkedIn

EDUCATION

B.S. Computer Engineering
Texas Tech University - Lubbock, Texas

Expected in: **Dec 2027**
GPA: **3.7**

PROJECTS & INVOLVEMENT

Raider Aerospace Society - Texas Tech University

Space Raiders - Avionics Team Lead

March 2025 – Present

- Successfully deployed Space Raiders first ever custom flight computer with working live telemetry and dynamic controls
- Developed and managed technical milestones and test schedules to drive project progress and accountability
- Directly managed interdisciplinary efforts of 15+ students across mechanical, electrical, and software
- Prepared avionics documentation for IREC 2026, consolidating block diagrams, wiring schematics, and design justifications
- Led integrated hardware and firmware development through competition launch, ensuring cross-domain functional verification
- Led cross-team integration for parachute deployment and aerodynamic braking systems, aligning with the mission profile

Space Raiders - APEX Flight Computer

August 2025 – Present

- Designed ARM Cortex-M7 based flight computer PCB with redundant sensor suites and signal integrity optimized 4-layer stackup
- Developed embedded firmware with interrupt based tasks and sensor fusion using Extended Kalman Filters
- Utilized real-time PID control loops using simulation-derived parameters to support aero-control surface actuation

Space Raiders - HORIZON Tracking Groundstation

August 2025 – Present

- Built a Raspberry Pi 5 groundstation platform with telemetry dashboard, 3D trajectory mapping, and web-based UI
- Developing multi-link RF communication architecture for redundant telemetry and video reception
- Prototyped pan-tilt actuation control for automated tracking using embedded control loops

Space Raiders - Airbrakes Flight Controls

August 2024 – July 2025

- Designed hardware, firmware, and control logic enabling automatic in-flight aero-braking to hit target altitude
- Enhanced sensor performance, timing discipline, and launch diagnostics through iterative hardware/firmware refinement
- Coordinated with avionics, recovery, and airbrakes teams to align deployment and mission objectives
- Created and executed integration test plans for firmware and hardware interactions
- Expanded firmware test coverage (timing checks, sensor sanity, on-pad diagnostics, unit testing) to reduce deployment risk
- Applied sensor filtering and state estimation (EWRA, Extended Kalman Filter) to improve embedded data quality

Firefly Aerospace - Cedar Park, Texas

Launch Vehicle Flight Software Intern

May 2026 – August 2026

- Supported advanced hardware-in-the-loop testing for major revision of flight software systems
- Worked closely with Avionics, Firmware, and Electrical Engineering to facilitate creation of universal vehicle configuration tool
- Integrated advanced CI/CD pipelines to support unit testing, cross-compilation, and package deployment

Personal Projects

Systems / Controls Engineer

October 2025 – Present

- Applied Linux development workflows across Ubuntu, Debian, Arch including remote SSH and service configuration
- Designed and prototyped mechanical/actuation components using CAD and 3D printing for embedded systems
- Prototyped real-time control loops in Python/C++ for custom control hardware
- Diagnosed night-before competition power brownouts; resolved via battery specification correction - **Pegasus (DBF Aircraft)**
- Developed and validated control system schematics for CO2 arming and ignition in a ground test rig - **Rolleron Ground Test Rig**

SKILLS

- **Software & Tools:** Git/GitHub, Matlab/Simulink, Altium, VS Code, Fusion360, OnShape, ESP-IDF, and CMake
- **Linux:** Ubuntu, Debian, Arch; CompTIA Linux+ certified; system/service configuration, user/permissions, remote SSH workflows
- **Windows:** toolchain usage, IDE debugging, environment configuration, batch scripting, file system permissions, CLI workflows
- **Networking Foundations:** IP addressing/subnetting, VLAN design, switch/interface configuration, and network segmentation
- **Embedded & Hardware:** Microcontrollers, RTOS, soldering, SPI/I²C/UART sensor integration, schematic/PCB design
- **Systems & Reliability:** Integration/test workflows, fault-isolation, safety interlocks & redundancy, basic signal conditioning
- **Languages:** C/C++, Python, Swift, JavaScript, Verilog/SystemVerilog, SQL, and Bash